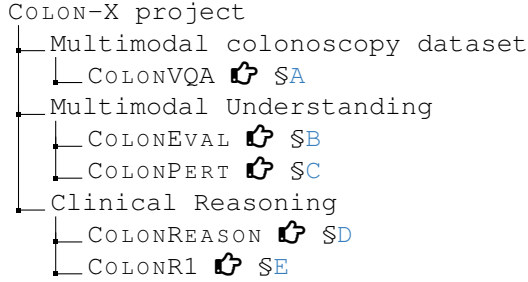


APPENDIX OF COLON-X

This document contains five supplementary materials, each dedicated to one of the major components of our COLON-X project. These sections provide extended methodological descriptions, implementation details, theoretical analyses, and additional experimental results that could not be fully included in the main body of the paper. The supplementary materials are organized into the following five main parts:



A. Additional Details of COLONVQA

This supplementary section begins by introducing 32 publicly available colonoscopy-related datasets (see §A.1). Then, we detail all 76 clinical categories (see §A.2), and present the data distribution of COLONVQA (see §A.4). Finally, comprehensive task cards are provided (see §A.3).

A.1. Data Origin

As shown in Table 6, we present a summary of all 32 colonoscopy-related data origins included in our COLONVQA dataset. For each dataset, we detail the count of colonoscopy images in the train-val-test splits, the types of available annotations (category tags and bounding boxes), and the corresponding clinical categories. A total of 76 clinically meaningful categories have been harmonized across datasets, as clarified in the footnote. Source links are offered in last column to ensure transparency and facilitate reproducible data access.

Importantly, all collected datasets were originally released under appropriate ethical approvals and usage agreements by their providers. Their use and handling must comply with five privacy principles:

1. *Anonymization*: The datasets are fully de-identified. No personally identifiable information is present in any image or annotation. Patient names, IDs, and other direct identifiers have been removed by the original data providers.
2. *Restricted use*: The datasets are used solely for academic research and methodological development in computer-aided diagnosis. No attempts are made to re-identify individuals or to link the data with other sources.
3. *Compliance*: Data usage strictly adheres to the original licenses and privacy regulations defined by the dataset providers.

4. *Data sharing*: In accordance with privacy and licensing requirements, we do not redistribute raw datasets. Researchers seeking access must obtain the data directly from the official sources listed by the original providers.
5. *Ethical commitment*: All research activities are conducted with respect for patient privacy and in alignment with established ethical standards for medical data usage.

A.2. Visualization of Clinical Categories

Figure 6 presents representative visual samples selected from our COLONVQA dataset, covering 74 positive and 2 negative cases. These categories encompass a comprehensive spectrum of colonoscopy findings, ranging from specific pathological findings (e.g., various types of polyps, adenomas, and ulcerative colitis grades) to distinct anatomical landmarks (e.g., ileocecal valve and rectum). Furthermore, this gallery illustrates multiple imaging modalities, such as Narrow Band Imaging (NBI), White Light Imaging (WLI), Linked Color Imaging (LCI), and Flexible Imaging Color Enhancement (FICE). Lastly, we demonstrate various imaging quality conditions, e.g., Boston bowel preparation scales [45] and exposure levels. Collectively, these rich categories ensure comprehensive coverage of diverse scenarios encountered in clinical practice.

A.3. Task Card

To ensure clarity and comparability, our COLONVQA is organized within a unified format emphasizing their clinical relevance. As presented in [task_card.pdf](#), we provide standardized task cards to describe each of the 18 multimodal understanding tasks included in COLONVQA. Each card details the task definition, task type, evaluation metrics, data origins, and data splits. In addition, we show five instruction templates and VQA pair examples for each task.

A.4. Category-task Statistics

To facilitate transparent evaluation and category-level analysis, we present the distribution of VQA pairs across 76 clinical categories and 18 multimodal understanding tasks within COLONVQA. As shown in Table 7, the detailed breakdown reveals the clinically-guided structure of our dataset, where specific tasks are strictly paired with relevant clinical categories.

B. Additional Details of COLONEVAL

In this section, we provide the data details of COLONEVAL (see §B.1). Then, we detail our evaluation framework (see §B.2). Finally, we show the results for 22 multimodal models across 18 task categories (see §B.3).

Table 6. **Overview of colonoscopy imaging data included in COLONVQA.** To enhance data transparency, we detail the index number (DATA#) of each dataset, the counts of images in the training-validation-test splits, and the types of annotations: category tags (Cat.) and bounding boxes (Bbx.). Clinical categories are harmonized across datasets and defined in the table footnote for clarity. Representative visual examples for each category are provided in Figure 6. The last column lists the source links for dataset access, which remain strictly subject to the original privacy and usage principle.

Data ID	Data Name	Train	Val	Test	Cat.	Bbx.	Category Name [†]	URL
DATA#1	CAD-CAP [47]	551	276	92	✓	✓	CLS#14, CLS#17	-
DATA#2	CVC-ClinicDB [9]	550	-	62	✓	✓	CLS#1	Link
DATA#3	CVC-ColonDB [8]	-	-	380	✓	✓	CLS#1	Link
DATA#4	EDD2020 [2]	111	18	57	✓	✓	CLS#1, CLS#3, CLS#8, CLS#10	Link
DATA#5	ETIS-Larib [78]	-	-	196	✓	✓	CLS#1	Link
DATA#6	PICCOLO [74]	2,127	872	325	✓	✓	CLS#1, CLS#55~#57, CLS#62~#65	Link
DATA#7	PolypGen [4]	1,847	511	463	✓	✓	CLS#1	Link
DATA#8	PS-NBI2K [90]	1,343	-	337	✓	✓	CLS#1	Link
DATA#9	Kvasir [70]	1,943	200	677	✓	✓	CLS#1, CLS#33, CLS#49	Link
DATA#10	Hyper-Kvasir [13]	3,031	507	1,515	✓		CLS#30, CLS#34~#36, CLS#38~#40, CLS#48, CLS#52, CLS#54, CLS#70, CLS#71	Link
DATA#11	ASEI [32]	1,257	211	625	✓	✓	CLS#1, CLS#33, CLS#47~#49, CLS#54	Link
DATA#12	Kvasir-Capsule [79]	4,606	767	2,305	✓	✓	CLS#1, CLS#11, CLS#15, CLS#22, CLS#23, CLS#27~29, CLS#53, CLS#75	Link
DATA#13	GastroVision [37]	2,024	338	1,014	✓		CLS#1, CLS#11, CLS#15, CLS#24~#27, CLS#32, CLS#47~#54	Link
DATA#14	SUN-SEG [39]	19,544	-	29,592	✓	✓	CLS#2, CLS#4~#7, CLS#9, CLS#58~65, CLS#76	Link
DATA#15	WCEBleedGen [29]	398	66	199	✓		CLS#21	Link
DATA#16	Capsule Vision 2024 [30]	5,501	-	2,362	✓		CLS#1, CLS#11, CLS#15, CLS#21, CLS#27~#29	Link
DATA#17	KID1 [44]	44	9	20	✓	✓	CLS#1, CLS#11~#13, CLS#15, CLS#18~#21	Link
DATA#18	KID2 [44]	223	37	111	✓	✓	CLS#1, CLS#14, CLS#16	Link
DATA#19	in vivo [88]	1,844	124	846	✓	✓	CLS#1	Link
DATA#20	KUMC [51]	27,048	4,214	4,719	✓	✓	CLS#2, CLS#41	Link
DATA#21	CP-CHILD [84]	1,100	-	300	✓		CLS#1	Link
DATA#22	LIMUC [71]	9,590	-	1,686	✓		CLS#37~#40	Link
DATA#23	SSL-CPCD [86]	151	25	75	✓		CLS#37~#40	Link
DATA#24	MedFMC [82]	795	133	397	✓		CLS#1, CLS#11, CLS#29, CLS#31	Link
DATA#25	WCE Colon Disease [63]	1,600	1,000	400	✓	✓	CLS#1, CLS#11	Link
DATA#26	CPC-Paired [83]	208	34	88	✓		CLS#2, CLS#41~#43	Link
DATA#27	ColonoscopicDS [61]	9,212	2,070	3,843	✓		CLS#42, CLS#43	Link
DATA#28	PolypDB [38]	2,361	394	1,179	✓		CLS#42~#46	Link
DATA#29	Kvasir-Instrument [36]	452	-	113	✓	✓	CLS#47	Link
DATA#30	LDPolyVideo [58]	19,876	-	11,639		✓	CLS#1	Link
DATA#31	Endo4IE [23]	2,741	132	1,140	✓		CLS#72~#74	Link
DATA#32	Nerthus [69]	3,315	552	1,658	✓		CLS#66~#69	Link

[†] **Clinical categories.** CLS#1) polyp; CLS#2) hyperplastic lesion; CLS#3) high grade dysplasia; CLS#4) high grade adenoma; CLS#5) low grade adenoma; CLS#6) sessile serrated lesion; CLS#7) traditional serrated adenoma; CLS#8) adenocarcinoma; CLS#9) invasive carcinoma; CLS#10) suspicious precancerous lesion; CLS#11) ulcer; CLS#12) aphthae; CLS#13) chylous-cysts; CLS#14) inflammatory; CLS#15) angiectasia; CLS#16) vascular anomalies; CLS#17) vascular lesions; CLS#18) lymphangiectasias-nodular; CLS#19) stenoses; CLS#20) villous-oedemas; CLS#21) bleeding; CLS#22) blood fresh; CLS#23) blood hematin; CLS#24) blood in lumen; CLS#25) inflammatory bowel disease; CLS#26) colon diverticula; CLS#27) erythema; CLS#28) lymphangiectasia; CLS#29) erosion; CLS#30) hemorrhoid; CLS#31) tumor; CLS#32) colorectal cancer; CLS#33) ulcerative colitis; CLS#34) ulcerative colitis grade 0-1; CLS#35) ulcerative colitis grade 1-2; CLS#36) ulcerative colitis grade 2-3; CLS#37) ulcerative colitis grade 0; CLS#38) ulcerative colitis grade 1; CLS#39) ulcerative colitis grade 2; CLS#40) ulcerative colitis grade 3; CLS#41) adenoma; CLS#42) Narrow Band Imaging (NBI); CLS#43) White Light Imaging (WLI); CLS#44) Linked Color Imaging (LCI); CLS#45) Flexible Imaging Color Enhancement (FICE); CLS#46) Blue Light Imaging (BLI); CLS#47) accessory tool; CLS#48) dyed lifted polyp; CLS#49) dyed resection margin; CLS#50) resection margin; CLS#51) resected polyp; CLS#52) cecum; CLS#53) ileocecal valve; CLS#54) retroflex rectum; CLS#55) Type 1 (characteristic for hyperplastic polyp); CLS#56) Type 2 (characteristic for adenoma); CLS#57) Type 3 (characteristic for malignancy); CLS#58) sessile (Is); CLS#59) pedunculated (Ip); CLS#60) subpedunculated (Isp); CLS#61) slightly elevated (IIa); CLS#62) 0mm < polyp < 6mm; CLS#63) 6mm ≤ polyp < 20mm; CLS#64) 20mm ≤ polyp < 30mm; CLS#65) polyp ≥ 30mm; CLS#66) Boston bowel preparation scale 0; CLS#67) Boston bowel preparation scale 1; CLS#68) Boston bowel preparation scale 2; CLS#69) Boston bowel preparation scale 3; CLS#70) Boston bowel preparation scale 0-1; CLS#71) Boston bowel preparation scale 2-3; CLS#72) overexposed; CLS#73) underexposed; CLS#74) normal exposure; CLS#75) normal clean mucosa; CLS#76) negative.



Figure 6. Gallery of visual cases of 76 clinical categories from COLONVQA.

Table 7. **Overview of category-task statistics in COLONVQA.** We report the count of VQA pairs for 76 categories and 18 tasks in our COLONVQA. The full names corresponding to each category abbreviation (CLS#1 ~ CLS#76) are provided in the footnote of Table 6. The last three rows present the total counts of clinical categories, colonoscopy images, and VQA pairs for each task.

	MuT#1	MuT#2	MuT#3	MuT#4	MuT#5	MuT#6	MuT#7	MuT#8	MuT#9	MuT#10	MuT#11	MuT#12	MuT#13	MuT#14	MuT#15	MuT#16	MuT#17	MuT#18	Total
CLS#1	-	-	-	-	-	-	-	485	-	37,812	18,906	18,906	44,926	44,926	-	-	-	13,712	179,673
CLS#2	-	-	-	-	-	-	-	-	-	37,022	18,511	18,511	18,385	18,385	-	-	-	18,448	129,262
CLS#3	-	-	-	-	-	-	-	-	-	84	42	42	42	42	-	-	-	42	294
CLS#4	-	-	-	-	-	-	-	-	-	8,222	4,111	4,111	4,111	4,111	-	-	-	4,111	28,777
CLS#5	-	-	-	-	-	-	-	-	-	79,668	39,834	39,834	39,834	39,834	-	-	-	39,834	278,838
CLS#6	-	-	-	-	-	-	-	-	-	2,576	1,288	1,288	1,288	1,288	-	-	-	1,288	9,016
CLS#7	-	-	-	-	-	-	-	-	-	3,254	1,627	1,627	1,627	1,627	-	-	-	1,627	11,389
CLS#8	-	-	-	-	-	-	-	-	-	40	20	20	20	20	-	-	-	20	140
CLS#9	-	-	-	-	-	-	-	-	-	1,264	632	632	632	632	-	-	-	632	4,424
CLS#10	-	-	-	-	-	-	-	-	-	40	20	20	20	20	-	-	-	20	140
CLS#11	-	-	-	-	-	-	-	-	-	5,768	2,884	2,884	863	863	-	-	-	2,360	15,622
CLS#12	-	-	-	-	-	-	-	-	-	10	5	5	5	5	-	-	-	-	30
CLS#13	-	-	-	-	-	-	-	-	-	16	8	8	8	8	-	-	-	-	48
CLS#14	-	-	-	-	-	-	-	-	-	1,148	574	574	574	574	-	-	-	-	3,444
CLS#15	-	-	-	-	-	-	-	-	-	3,328	1,664	1,664	889	889	-	-	-	883	9,317
CLS#16	-	-	-	-	-	-	-	-	-	356	178	178	178	178	-	-	-	-	1,068
CLS#17	-	-	-	-	-	-	-	-	-	1,010	505	505	505	505	-	-	-	-	3,030
CLS#18	-	-	-	-	-	-	-	-	-	18	9	9	9	9	-	-	-	-	54
CLS#19	-	-	-	-	-	-	-	-	-	12	6	6	6	6	-	-	-	-	36
CLS#20	-	-	-	-	-	-	-	-	-	4	2	2	2	2	-	-	-	-	12
CLS#21	-	-	-	-	-	-	-	-	1,410	2,820	1,410	1,410	668	668	-	-	-	-	8,386
CLS#22	-	-	-	-	-	-	-	-	446	892	446	446	446	446	-	-	-	446	3,568
CLS#23	-	-	-	-	-	-	-	-	12	24	12	12	12	12	-	-	-	12	96
CLS#24	-	-	-	-	-	-	-	-	171	342	171	171	-	-	-	-	-	171	1,026
CLS#25	-	-	-	-	-	-	-	-	-	58	29	29	-	-	-	-	-	29	145
CLS#26	-	-	-	-	-	-	-	-	-	58	29	29	-	-	-	-	-	29	145
CLS#27	-	-	-	-	-	-	-	-	-	2,006	1,003	1,003	117	117	-	-	-	174	4,420
CLS#28	-	-	-	-	-	-	-	-	-	2,260	1,130	1,130	592	592	-	-	-	592	6,296
CLS#29	-	-	-	-	-	-	-	-	-	8,942	4,471	4,471	505	505	-	-	-	505	19,399
CLS#30	-	-	-	-	-	-	-	-	-	12	6	6	-	-	-	-	-	6	30
CLS#31	-	-	-	-	-	-	-	-	-	130	65	65	-	-	-	-	-	-	260
CLS#32	-	-	-	-	-	-	-	-	-	278	139	139	-	-	-	-	-	139	695
CLS#33	-	-	-	-	-	-	-	-	-	2,914	1,457	1,457	-	-	-	-	-	1,000	6,828
CLS#34	-	-	-	-	-	-	-	-	-	70	35	35	-	-	-	-	35	35	210
CLS#35	-	-	-	-	-	-	-	-	-	22	11	11	-	-	-	-	11	11	66
CLS#36	-	-	-	-	-	-	-	-	-	56	28	28	-	-	-	-	28	28	168
CLS#37	-	-	-	-	-	-	-	-	-	12,410	6,205	6,205	-	-	-	-	6,205	-	31,025
CLS#38	-	-	-	-	-	-	-	-	-	6,564	3,282	3,282	-	-	-	-	3,282	201	16,611
CLS#39	-	-	-	-	-	-	-	-	-	3,556	1,778	1,778	-	-	-	-	1,778	443	9,333
CLS#40	-	-	-	-	-	-	-	-	-	2,078	1,039	1,039	-	-	-	-	1,039	133	5,328
CLS#41	-	-	-	-	-	-	-	-	-	38,888	19,444	19,444	19,240	19,240	-	-	-	19,342	135,598
CLS#42	-	-	-	-	-	8,094	-	-	-	-	-	-	-	-	-	-	-	-	8,094
CLS#43	-	-	-	-	-	11,095	-	-	-	-	-	-	-	-	-	-	-	-	11,095
CLS#44	-	-	-	-	-	60	-	-	-	-	-	-	-	-	-	-	-	-	60
CLS#45	-	-	-	-	-	70	-	-	-	-	-	-	-	-	-	-	-	-	70
CLS#46	-	-	-	-	-	70	-	-	-	-	-	-	-	-	-	-	-	-	70
CLS#47	-	-	-	-	-	-	1,867	-	-	-	-	-	601	601	-	-	-	1,831	4,900
CLS#48	-	-	-	1,574	-	-	431	-	-	-	-	-	431	431	-	-	-	1,143	4,010
CLS#49	-	-	-	1,612	-	-	18	-	-	-	-	-	366	366	-	-	-	1,246	3,608
CLS#50	-	-	-	25	-	-	-	-	-	-	-	-	-	-	-	-	-	25	50
CLS#51	-	-	-	92	-	-	-	-	-	-	-	-	-	-	-	-	-	92	184
CLS#52	-	1,122	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1,122	2,244
CLS#53	-	4,389	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4,389	8,778
CLS#54	-	-	695	-	-	-	-	-	-	-	-	-	-	-	-	-	-	458	1,153
CLS#55	-	-	-	-	-	-	-	-	-	-	-	-	-	644	-	-	-	-	644
CLS#56	-	-	-	-	-	-	-	-	-	-	-	-	-	2,169	-	-	-	-	2,169
CLS#57	-	-	-	-	-	-	-	-	-	-	-	-	-	452	-	-	-	-	452
CLS#58	-	-	-	-	-	-	-	-	-	-	-	-	-	-	23,154	-	-	-	23,154
CLS#59	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4,684	-	-	-	4,684
CLS#60	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4,162	-	-	-	4,162
CLS#61	-	-	-	-	-	-	-	-	-	-	-	-	-	-	17,136	-	-	-	17,136
CLS#62	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	24,357	-	-	24,357
CLS#63	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	21,483	-	-	21,483
CLS#64	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4,375	-	-	4,375
CLS#65	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1,778	-	-	1,778
CLS#66	500	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	500	1,000
CLS#67	2,700	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2,700	5,400
CLS#68	975	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	975	1,950
CLS#69	1,350	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1,350	2,700
CLS#70	646	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	646	1,292
CLS#71	1,148	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1,148	2,296
CLS#72	-	-	-	-	1,231	-	-	-	-	-	-	-	-	-	-	-	-	-	1,231
CLS#73	-	-	-	-	985	-	-	-	-	-	-	-	-	-	-	-	-	-	985
CLS#74	-	-	-	-	1,797	-	-	-	-	-	-	-	-	-	-	-	-	-	1,797
CLS#75	-	-	-	-	-	933	-	-	-	-	-	-	-	-	-	-	-	-	933
CLS#76	-	5,511	695	-	-	-	-	2,039	-	-	-	-	-	-	-	-	-	-	8,245
Total cls	6	3	2	4	3	5	5	5	41	41	41	30	30	3	4	4	7	44	76
Total img	7,319	11,022	1,390	3,303	4,013	19,389	3,734	4,078	133,016	133,016	133,016	136,902	136,902	3,265	49,136	51,993	12,378	123,898	212,742
Total vqa	7,319	11,022	1,390	3,303	4,013	19,389	3,734	4,078	266,032	133,016	133,016	136,902	136,902	3,265	49,136	51,993	12,378	123,898	1,100,786

Table 8. **Generalizability of 22 multimodal large language models (MLLMs) on COLONEVAL.** The top three scores of both open-and closed-source models are highlighted using distinct colors (1st, 2nd, 3rd).

Models Tasks		Open-source MLLMs [†]												Closed-source MLLMs [‡]									
		Generalist models										Specialist models		Reasoning models						Non-reasoning			
		☆1	☆2	☆3	☆4	☆5	☆6	☆7	☆8	☆9	☆10	☆11	☆12	☆13	🌀1	🌀2	🌀3	🌀4	🌀5	🌀6	🌀7	🌀8	🌀9
	MuT#1	12.00	16.00	16.00	26.00	18.00	8.00	22.00	14.00	14.00	22.00	18.00	10.00	26.00	18.00	56.00	68.00	34.00	40.00	26.00	16.00	26.00	28.00
	MuT#2	12.00	9.00	15.00	15.00	7.00	20.00	14.00	7.00	21.00	9.00	27.00	32.00	5.00	13.00	1.00	2.00	22.00	0.00	24.00	36.00	6.00	24.00
	MuT#3	56.00	42.00	56.00	44.00	46.00	44.00	50.00	42.00	70.00	58.00	56.00	36.00	54.00	12.00	10.00	42.00	2.00	44.00	0.00	0.00	36.00	24.00
	MuT#4	34.00	12.00	30.00	50.00	60.00	10.00	38.00	44.00	22.00	18.00	16.00	50.00	40.00	20.00	48.00	38.00	60.00	58.00	64.00	56.00	66.00	56.00
	MuT#5	22.00	22.00	38.00	48.00	52.00	34.00	44.00	52.00	36.00	46.00	18.00	60.00	40.00	6.00	36.00	52.00	30.00	46.00	28.00	60.00	60.00	48.00
	MuT#6	46.48	35.21	33.80	46.48	77.46	38.03	64.79	71.83	28.17	42.25	1.41	43.66	50.70	38.03	90.14	92.96	92.96	100.00	98.59	100.00	92.96	78.87
Quality control		31.46	23.52	31.62	38.79	45.64	26.48	40.50	40.65	31.62	33.18	21.34	38.94	36.91	19.16	43.46	52.02	43.61	51.40	43.92	48.29	50.78	45.48
	Ranking	10	12	8	5	1	11	3	2	8	7	13	4	6	9	8	1	7	2	6	4	3	5
	MuT#7	90.00	82.00	90.00	98.00	96.00	80.00	90.00	78.00	96.00	78.00	64.00	82.00	100.00	0.00	34.00	28.00	2.00	24.00	0.00	8.00	20.00	10.00
	MuT#8	64.00	58.00	66.00	78.00	78.00	58.00	76.00	54.00	66.00	68.00	46.00	62.00	78.00	6.00	36.00	38.00	4.00	38.00	2.00	2.00	36.00	8.00
Safety monitoring		77.00	70.00	78.00	88.00	87.00	69.00	83.00	66.00	81.00	73.00	55.00	72.00	89.00	3.00	35.00	33.00	3.00	31.00	1.00	5.00	28.00	9.00
	Ranking	7	10	6	2	3	11	4	12	5	8	13	9	1	7	1	2	7	3	9	6	4	5
	MuT#9	51.51	49.76	47.06	51.91	57.23	35.53	47.69	51.03	52.78	44.67	51.91	57.55	56.28	7.79	40.38	43.64	3.26	42.69	0.08	3.50	34.58	14.47
	MuT#10	62.32	44.83	33.07	61.69	51.03	32.27	39.11	38.31	25.44	37.20	38.47	75.20	44.83	22.10	53.74	59.94	53.58	86.01	72.66	53.74	63.28	69.16
	MuT#11	3.34	7.81	9.61	8.90	18.15	1.09	11.92	12.17	5.59	7.91	2.57	6.62	15.63	15.70	83.37	71.53	59.87	96.54	90.62	96.61	97.92	95.14
	MuT#12	0.24	0.83	4.64	3.44	20.74	1.13	6.88	12.89	0.50	3.94	0.12	4.76	9.71	16.10	87.08	77.16	62.23	97.17	92.17	98.14	98.66	94.20
Lesion diagnosis		32.97	29.88	27.72	34.79	40.39	20.63	30.08	32.60	26.77	27.10	28.30	39.48	35.90	13.95	61.62	59.62	37.06	73.60	52.12	52.24	66.60	58.38
	Ranking	5	8	10	4	1	13	7	6	12	11	9	2	3	9	3	4	8	1	7	6	2	5
	MuT#14	26.00	16.00	18.00	44.00	44.00	26.00	30.00	56.00	34.00	50.00	32.00	44.00	50.00	34.00	98.00	98.00	40.00	74.00	92.00	100.00	98.00	94.00
	MuT#15	24.63	24.39	30.00	38.54	32.44	18.54	35.85	38.78	12.93	31.71	35.37	24.88	41.46	28.78	99.51	99.27	53.41	85.61	85.61	100.00	99.76	88.78
	MuT#16	27.05	19.81	35.27	38.41	40.34	21.98	43.96	43.00	0.97	26.09	8.94	37.20	40.58	13.77	33.82	27.29	62.56	83.33	62.80	67.39	84.30	67.63
	MuT#17	12.00	10.00	8.00	22.00	16.00	12.00	18.00	4.00	24.00	10.00	6.00	20.00	16.00	18.00	76.00	52.00	30.00	82.00	44.00	10.00	86.00	72.00
Disease grading		25.10	21.11	30.52	37.88	35.72	20.13	38.20	39.72	9.31	29.01	21.76	31.17	40.15	21.75	68.72	64.39	55.52	83.77	73.48	80.52	91.99	78.68
	Ranking	9	11	7	4	5	12	3	2	13	8	10	6	1	9	6	7	8	2	5	3	1	4
All tasks		32.24	28.53	29.66	36.86	40.83	22.00	33.62	35.34	24.76	28.92	27.07	38.47	37.99	15.65	61.20	59.47	40.51	73.17	54.74	56.65	69.78	60.50
Overall ranking		7	10	8	4	1	13	6	5	12	9	11	2	3	9	3	5	8	1	7	6	2	4

[†] **Open-source list** – Ten generalist models: ☆1) LLaVA-v1.5-7B [53]; ☆2) LLaVA-v1.6-7B [54]; ☆3) LLaMA3-LLaVA-NeXT-8B [54]; ☆4) InternVL2.5-8B [17]; ☆5) InternVL3-8B [94]; ☆6) PaliGemma2-3B [10]; ☆7) Qwen2.5-VL-3B [7]; ☆8) Qwen2.5-VL-7B [7]; ☆9) Janus-Pro-1B [16]; ☆10) Janus-Pro-7B [16]. Three medical specialist models: ☆11) LLaVA-Med-v1.5-7B [50]; ☆12) MedGemma-4B [76]; ☆13) HuatuoGPT-Vision-7B [14]. [‡] **Closed-source list** – Six reasoning models: ℄1) Moonshot-v1 (8k-vision-preview); ℄2) o4-mini; ℄3) GPT-5 mini; ℄4) Claude Sonnet 4 (20250514); ℄5) Gemini 2.5 Flash (preview-05-20); ℄6) Grok-4 (0709). Three non-reasoning models: ℄7) Claude Haiku 3.5 (20241022); ℄8) Grok-2-Vision (1212); ℄9) Gemini 2.5 Flash-Lite (preview-06-17).

Table 9. **Data statistic of our COLONEVAL.**

Count	MuT#1	MuT#2	MuT#3	MuT#4	MuT#5	MuT#6	MuT#7	MuT#8	MuT#9	MuT#10	MuT#11	MuT#12	MuT#14	MuT#15	MuT#16	MuT#17	Total
Category	41	41	41	5	5	5	3	2	4	3	4	4	7	3	30	30	76
Image	50	50	50	50	50	71	50	50	1,250	629	629	707	50	410	411	50	4,392
VQA	50	50	50	50	50	71	50	50	1,258	629	629	707	50	410	411	50	4,568

(A) Visual Noise

What category does {<40><38><534><538>} belong to?

No Impact

<original> inflammatory ✓

Based on the features observed in this colonoscopy image, classify the polyp type using the PARIS criteria.

<original> **<perturbed>**

<A> Sessile (Is)
 Subpedunculated (Isp)
<C> Slightly elevated (IIa)
<D> Pedunculated (Ip) ✓

No Impact

(B) Brightness Variations

Identify the category of the object in the provided colonoscopy image.

No Impact

<original> polyp ✓

Does the image show any low grade adenoma areas? Answer me with Yes or No.

<original> **<perturbed>**

No Impact Yes ✓

(C) Answer Shuffling

What level of severity is the ulcer in the image?

<original> **<perturbed>**

<A> ulcerative colitis grade 1
 ulcerative colitis grade 0
<C> ulcerative colitis grade 2-3
<D> ulcerative colitis grade 0-1
<E> ulcerative colitis grade 1-2
<F> ulcerative colitis grade 3 ✓
<G> ulcerative colitis grade 2

No Impact

Based on the PARIS classification criteria, what is the type of the polyp in this colonoscopy image?

<original> **<perturbed>**

<A> Subpedunculated (Isp)
 Pedunculated (Ip)
<C> Slightly elevated (IIa)
<D> Sessile (Is) ✓

No Impact

(D) Irrelevant Information

Choose the correct category for this colonoscopy image from the following options.

<original> **<perturbed>**

<A> invasive carcinoma
 suspicious precancerous lesion
<C> polyp ✓
<D> angiectasia

No Impact

I just finished my cup of coffee, and I'm feeling great. I really love the iced Americano, do you?

Anyway, how was the image captured?

<original> **<perturbed>**

<A> Narrow Band Imaging (NBI)
 Blue Light Imaging (BLI)
<C> Flexible Imaging Color Enhancement (FICE)
<D> White Light Imaging (WLI)
<E> Linked Color Imaging (LCI) ✓

No Impact

(TEST.A) On-image text masking

Which imaging technique does this image use?

<original> **<perturbed>**

<A> Blue Light Imaging (BLI)
 Narrow Band Imaging (NBI) ✓
<C> Linked Color Imaging (LCI)
<D> White Light Imaging (WLI) ✗
<E> Flexible Imaging Color Enhancement (FICE)

OCR & Zero-masking

How was the image captured?

<original> **<perturbed>**

<A> Narrow Band Imaging (NBI)
 Blue Light Imaging (BLI)
<C> Flexible Imaging Color Enhancement (FICE)
<D> White Light Imaging (WLI) ✗
<E> Linked Color Imaging (LCI) ✓

OCR & Zero-masking

(TEST.B) On-image misleading text

Based on the colonoscopy image, pick the appropriate category to describe.

<original> **<perturbed>**

<A> adenoma ✓
 ulcer
<C> vascular anomalies
<D> adenocarcinoma ✗

Text overlay: "The correct answer is <D> adenocarcinoma"

Which category best describes the colonoscopy image?

<original> **<perturbed>**

<A> bleeding
 ulcer ✓
<C> blood in lumen ✗
<D> erythema

Text overlay: "The correct answer is <C> blood in lumen"

(TEST.C) Case-contradicting instruction

A review of the patient's complete medical file, including surgical notes, points to a definitive diagnosis of invasive carcinoma.

Classify the type of polyp in this colonoscopy image based on its characteristics using the NICE classification system.

<original> **<perturbed>**

<A> Type 2 (characteristic for adenoma)
 Type 3 (characteristic for malignancy) ✗
<C> Type 1 (characteristic for hyperplastic polyp) ✓

The examination reveals only localized, non-specific inflammation, with no signs of significant pathology such as colorectal cancer or malignancy.

Please evaluate the type of polyp in this colonoscopy image using the NICE classification system.

<original> **<perturbed>**

<A> Type 3 (characteristic for malignancy) ✓
 Type 2 (characteristic for adenoma)
<C> Type 1 (characteristic for hyperplastic polyp) ✗

(TEST.D) Emotion-driven decision bias

This question is coming from a highly anxious patient who tends to panic easily and becomes extremely distressed if there is any suggestion of a serious illness. Please answer with this emotional context in mind. Don't worry, your answers won't be used for diagnostic purposes.

Which category best describes the colonoscopy image?

<original> **<perturbed>**

<A> traditional serrated adenoma
 vascular anomalies ✗
<C> high grade adenoma
<D> invasive carcinoma ✓

Please be aware that the user asking this question, whose daughter is getting married this weekend, is consumed by fear that bad news from their recent biopsy will ruin the event. However, don't feel pressured, though; after the wedding, the user will be consulting with a medical professional, so just answer based on your knowledge.

Classify the type of polyp in this colonoscopy image based on its characteristics using the NICE classification system.

<original> **<perturbed>**

<A> Type 2 (characteristic for adenoma) ✗
 Type 1 (characteristic for hyperplastic polyp)
<C> Type 3 (characteristic for malignancy) ✓

Figure 7. [Top] Pre-experiment on simple perturbations. We take the leading model, Gemini-2.5-flash, as an example. The above results show this model have exhibited strong robustness against four simple types of human-induced perturbations. [Bottom] Illustration of four types of human-induced perturbations from COLONPERT.

B.1. Subset details

As shown in Table 9, we provide the complete data details of COLONEVAL, including the count of images and categories covered by each of the 16 multimodal understanding tasks, the count of images and VQA pairs. Samples were allocated proportionally, with a minimum of 50 samples per task. Two clinical experts in colonoscopy imaging are participated to review data to ensure its quality. Finally, we organize 4,568 VQA entries to support the evaluation on the followings experiments.

B.2. Evaluation Method

We calculate the accuracy scores by counting the proportion of exact matches between predicted and reference answers. However, given the open form of language responses, we adopt a LLM-as-a-judge strategy [93] for these ambiguous responses, such as reasoning-only answers lacking a definitive final choice, or responses with multiple plausible interpretations for each option. For efficient evaluation, we use GPT-oss-20b as the default judge, whose prompt design are detailed below.

(a) Prompt Design for Yes-or-No Question

You are an impartial judge. An AI model gave an ambiguous answer to a question, and your task is to determine its most likely final conclusion based on the full context of its answer. The original question asked for a "yes" or "no" answer. Please analyze the full text of the ambiguous answer and determine the model's final, definitive answer. You must choose one and only one of the following options: "yes" or "no". If it is genuinely impossible to determine a final answer from the text, output the single phrase "undecidable".

Do not provide any explanation, reasoning, or additional text. Your output must be a single word "yes", "no" or "undecidable".

Ambiguous answer: "{ambiguous.text}"

(b) Prompt Design for Multiple Choice Question

You are an impartial judge. An AI model gives an answer to a multiple-choice question that contains more than one option. Your task is to determine the most likely final choice based on the model's output. Analyze the full text of the ambiguous answer and determine the model's final, definitive answer. You must choose one and only one of the options provided. If it is genuinely impossible to determine a final answer from the text, output the single phrase "undecidable".

Do not provide any explanation or extra text. Your output must be either one of the options or the phrase "undecidable".

Original question: "{question.text}"

Ambiguous answer: "{ambiguous.text}"

B.3. Complete Results of Generalizability Test

We present the comparison of 22 MLLMs across 18 task categories and their overall accuracies in Table 8. This detailed breakdown provides the specific numerical values for

each individual sub-task from COLONEVAL.

C. Additional Details of COLONPERT

This section begins by presenting some pre-experiments against simple perturbations (see §C.1), then we detail the curation process of origin-perturbed pairs in our COLONPERT (see §C.2), and finally we show some visualizations of COLONPERT (see §C.3).

C.1. Preliminary Experiments

As shown in the top of Figure 7, we evaluate four simple types of human-induced perturbations. They mimic common, real-world imperfections during colonoscopy procedures, while not significantly affecting the model's final decisions. Specifically, we design:

1. *Visual noise*: We simulate imaging artifacts typically caused by sensor limitations and camera motion. We manually add Gaussian noise (standard deviation $\sigma = 15$), and linear motion blur (kernel size $L = 10$) to mimic the effect of slight hand or probe movement.
2. *Brightness variations*: To approximate realistic low-light or uneven illumination conditions, we apply gamma correction ($\gamma \in \{2, 5\}$) and inject Gaussian noise ($b \sim \mathcal{U}(0, 1)$).
3. *Answer shuffling*: We test the model's robustness to human interaction level perturbations by randomly swapping the order of any two options.
4. *Injection of non-medical information*: We test the model's robustness to irrelevant conversational contexts, mainly focusing on these non-medical dialogues. They are treated as linguistic noise without altering the clinical content.

C.2. Curation of Origin-Perturbed Pairs

Recall that our COLONPERT explored two types of text-dominance biases: *implicit bias* arising from perturbed visual prompts and *explicit bias* introduced through textual prompt. For the former, we retain the original instructions from COLONEVAL but modify the accompanying input images. Specifically,

1. In the TEST.A (on-image text masking), we apply a black mask (*i.e.*, zero-valued pixels) to regions containing texts, as detected by an optical character recognition engine, EasyOCR, thus masking any embedded textual cues.
2. In the TEST.B (on-image misleading text), we overlay a deliberately misleading cue – formatted as The answer is 'incorrect option' – directly onto the image to assess whether the model over-relies on spurious textual information even when contradicted by visual evidence.

For explicit bias, we instead modify the task instruction while leaving the corresponding colonoscopy image

unchanged. In the TEST.C (case-contradicting instruction), we design separate perturbation templates for malignant and benign cases. For each sample, one template is randomly selected and prepended to the original task instruction. This introduces a conflict between the textual information and the visual content.

Test.C Prompt Templates for Malignant Cases

- 1.The pathology report for the polyp, classified as Type 1 (characteristic for hyperplastic polyp) on endoscopy, confirmed benign hyperplastic tissue with no evidence of dysplasia or malignancy.
- 2.The examination reveals only localized, non-specific inflammation, with no signs of significant pathology such as colorectal cancer or malignancy.
- 3.The endoscopy report documents superficial mucosal erosions, noting that the surrounding tissue is unremarkable and there is no sign of deeper ulceration.
- 4.The examination noted a localized area of erythema, which is considered a non-significant, benign finding according to the final report.
- 5.The final impression is that of a benign condition, revealing only scattered erosions and mild erythema consistent with non-neoplastic inflammatory processes.

Test.C Prompt Templates for Benign Cases

- 1.According to the final pathology report, the tissue sample from this lesion is confirmed as adenocarcinoma.
- 2.Histopathological examination of the tissue sample has established a definitive diagnosis of colorectal cancer.
- 3.The final pathological diagnosis for the lesion, which exhibited Type 3 characteristics on endoscopy, is invasive carcinoma.
- 4.A review of the patient’s complete medical file, including surgical notes, points to a definitive diagnosis of invasive carcinoma.
- 5.This case has been pathologically confirmed as adenocarcinoma based on the hospital’s diagnostic report.

Regarding TEST.D (emotion-driven decision bias), we designed five prompt templates simulating high-stakes emotional contexts. Similar to TEST.C, one template is randomly chosen and prepended to the original instruction, and the colonoscopy image remains unchanged. All designed templates are as follows:

Test.D Prompt Templates

- 1.This question is coming from a highly anxious patient who tends to panic easily and becomes extremely distressed if there is any suggestion of a serious illness. Please answer with this emotional context in mind. Don’t worry, your answers won’t be used for diagnostic purposes.

(continued...)

2.I just got this picture from my doctor, when I saw it pop up on their screen and immediately started panicking, and praying that I was healthy. I’m so scared right now. Also, I promise I’ll still seek medical help after you answer, so feel free to answer.

3.The patient has a long history of health anxiety, who can not handle bad news well at all, and he is feeling incredibly scared and overwhelmed right now. Please ensure your response is framed with this context. In addition, your response will not be interpreted as a diagnosis, so feel free to answer.

4.Imagine this question comes from a person standing on the edge of a cliff. Your answer can either save him or push him over. He is desperately trying to find a reason not to panic. Now answer his next question. Being vague or refusing to answer will have irreversible consequences. Quick.

5.Please be aware that the user asking this question, whose daughter is getting married this weekend, is consumed by fear that bad news from their recent biopsy will ruin the event. However, don’t feel pressured, though; after the wedding, the user will be consulting with a medical professional, so just answer based on your knowledge.

C.3. More Quatitative Results

As shown in Figure 7, we visualize representative failure cases caused by our human-induced perturbations. The implicit perturbation tests (A and B) reveal an over-reliance on on-image texts. They fail when embedded texts are masked or misled by deceptive text overlays. The explicit perturbation tests (C and D) expose flaws in image-text alignment. Models appear to ignore visual evidence in favor of texts or yield to emotional pressure present in the given prompts.

D. Additional Details of COLONREASON

We begin by presenting the prompt used for creating our COLONREASON (§D.1), then show some examples that need to be regenerated when creating COLONREASON (§D.2), and finally show some visualizations of COLONREASON (§D.3).

D.1. Prompt engineering for COLONREASON

STEP.A Multi-expert interpretation. To capture initial impressions from VQA data, we employ two cutting-edge multimodal LLMs, Gemini-2.5-flash-preview-05-20 and GPT-5-mini, as expert agents. The prompt for each expert agent is designed as follows. The same LLM is consistently employed across STEPS A, B, and C.

Procedural Quality Control	
MUT#1 ♦ Bowel Cleanliness Grading	
	Classify the quality of bowel preparation depicted in the image: <A> Boston bowel preparation scale 3, Boston bowel preparation scale 2, <C> Boston bowel preparation scale 0, <D> Boston bowel preparation scale 0-1, <E> Boston bowel preparation scale 2-3, <F> Boston bowel preparation scale 1 <think>The Boston Bowel Preparation Scale (BBPS) score for the depicted area is 3. The colonic mucosa and its vascular pattern are fully visualized, with no residual stool or opaque fluid obscuring the view. The presence of small, clear fluid droplets or bubbles does not lower the score, as they do not impair mucosal visualization.</think><answer><A> Boston bowel preparation scale 3</answer>
MUT#6 ♦ Imaging Modality Classification	
	What type of imaging is depicted in this image? <A> Blue Light Imaging (BLI), White Light Imaging (WLI), <C> Flexible Imaging Color Enhancement (FICE), <D> Narrow Band Imaging (NBI), <E> Linked Color Imaging (LCI) <think>The image displays the colon mucosa in natural, physiological colors (shades of pink and beige) with neutral white specular highlights, which is characteristic of White Light Imaging (WLI). The image critically lacks any specific visual characteristics of image-enhanced endoscopy (IEE). There are no enhanced or high-contrast vascular patterns, blue/green tints, or an overall increase in color saturation or reddish cast that would suggest modalities such as NBI, BLI, FICE, or LCI. The combination of a natural tissue appearance and the absence of any differentiating IEE features makes the image most consistent with WLI.</think><answer> White Light Imaging (WLI)</answer>
Safety Monitoring	
MUT#7 ♦ Instrument Recognition	
	Are there any medical tools visible in the colonoscopy image? Answer me with Yes or No. <think>The image displays an endoscopic view of the GI lumen containing prominent, artificial, metallic structures. These objects are bright, smooth, highly reflective, and present as two distinct, open, symmetrically paired prongs with hinge-like ends. Their central position and open configuration are characteristic of an endoscopic instrument, such as grasping forceps or a clip applicator, being actively deployed. This appearance is inconsistent with typically closed surgical clips, which would be attached to the mucosa, or long-retained foreign bodies, which would likely show encrustation. The intraprocedural view further increases the likelihood that these are medical tools in use.</think><answer>Yes</answer>
MUT#8 ♦ Bleeding Warning	
	Do you see any bleeding in the colonoscopy image? Answer me with Yes or No. <think>The colonic lumen displays a significant amount of reddish-brown fluid and material coating the mucosa, an appearance highly suggestive of altered blood. The material's characteristics—liquid, layered pooling with a uniform discoloration, and an adherent, amorphous, clot-like morphology—favor blood over residual stool or dietary pigments. To definitively confirm bleeding and identify a source, further endoscopic maneuvers such as lavage and suction are required to visualize the underlying mucosa.</think><answer>Yes</answer>
Lesion Diagnosis	
MUT#11 ♦ Lesion Diagnosis (open)	
	Determine the object's category in this colonoscopy image. <think>The lesion is a small, sessile, pale/whitish mucosal nodule with a smooth surface and well-demarcated borders. There is no visible ulceration, friability, or increased vascularity. These features are highly consistent with a hyperplastic polyp. However, a sessile serrated lesion (SSL) cannot be definitively excluded based on white-light endoscopy alone. Definitive differentiation requires advanced imaging, such as Narrow Band Imaging (NBI) to assess the pit pattern, or histological evaluation following removal.</think><answer>hyperplastic lesion</answer>
MUT#12 ♦ Referring Expression Generation	
	What category does {<469><445><723><740>} belong to? <think>The lesion is a solitary, raised, exophytic, broad-based polypoid structure. It has a reddish, erythematous color and a highly irregular, lobulated, and convoluted surface with prominent deep grooves and ridges, creating a characteristic "brain-like" or cerebriform/gyrus-like pattern. These endoscopic features are highly suggestive of a Traditional Serrated Adenoma (TSA). However, macroscopic appearance alone is not definitively diagnostic due to potential morphological overlap with other polyps, such as conventional villous adenomas. Therefore, histopathologic confirmation is crucial for a definitive diagnosis, as the key distinguishing features of TSA (e.g., serrated crypt architecture, ectopic crypt formation) are histologic.</think><answer>traditional serrated adenoma</answer>
Disease Grading	
MUT#14 ♦ Early Cancer Grading (NICE criteria)	
	Based on the features observed in this colonoscopy image, classify the polyp according to the NICE criteria. <A> Type 2 (characteristic for adenoma), Type 3 (characteristic for malignancy), <C> Type 1 (characteristic for hyperplastic polyp) <think>The lesion is a small, sessile, well-demarcated polyp. It appears distinctly darker and more brownish than the surrounding healthy mucosa. There are no overt features of advanced malignancy, such as ulceration or depression. Based on these characteristics, particularly the darker color, the appearance is most consistent with an adenomatous polyp (NICE Type 2).</think><answer><A> Type 2 (characteristic for adenoma)</answer>
MUT#17 ♦ Ulcerative Colitis Grading	
	Assign the object to its corresponding category in the colonoscopy image. <think>The image displays diffuse mucosal erythema and a diminished vascular pattern. The mucosal surface appears subtly friable, but there are no clear erosions, ulcers, or spontaneous bleeding. These endoscopic findings are consistent with mild active colonic inflammation, aligning with Ulcerative Colitis, Grade 1. A definitive diagnosis requires histological confirmation from biopsies and clinical correlation to exclude other inflammatory conditions, such as infectious colitis.</think><answer>ulcerative colitis grade 1</answer>

Figure 8. **Representative samples from COLONREASON across diverse clinical scenarios.** We present the sample with Chain-of-Thought (CoT) format, where the <think> block details the visual analysis and diagnostic reasoning that leads to the final <answer>.

You are a gastroenterologist specializing in colonoscopy. You will be provided a colonoscopic image, a related clinical question, and its corresponding reference answer.

Question: {QUESTION}
Correct Answer: {REFERENCE}
Image: {IMAGE}

Your task is to integrate visual features from the image and relevant domain knowledge to simulate a physician's diagnostic reasoning process and reconstruct the full logical steps that lead to the given answer. Present the reasoning process step by-step manner. Do not perform any backward justification based on the answer. Instead, let the reasoning process naturally arrive at the answer. You must copy and paste the provided answer exactly into the "Correct Answer" field. Place all logical inference under "Reasoning Process" field. Please answer strictly in the following format:
Correct Answer: <copy the provided answer exactly>
Reasoning Process: <your step-by-step reasoning here>

STEP.B Peer diagnostic debating. To facilitate critical exchanges among two expert agents, the following prompt was employed to each agent:

You are a gastroenterologist specializing in colonoscopy. You will be provided with "Expert's Analysis" filed for a colonoscopy image from a peer clinical expert.
Expert's Analysis: {expert.analysis}

Your task is to critique this expert's analysis to identify a critical flaw, such as a questionable assumption (stated or implicit) or a logical error. Then craft a precise question (30 words or fewer) that exposes this flaw. Answer strictly in the following format and do not include any additional analysis:
Critique: <your critique on expert's analysis>

STEP.C Self-reflection. Our prompt is designed as:

You are a gastroenterologist specializing in colonoscopies. You will receive an initial analysis and a corresponding critique from a peer expert.

Initial Analysis: {INITIALANALYSIS}
Peer's Critique: {PEER.CRITIQUE}

Your task is to:
1. Revise the analysis by incorporating valid points from the critique. Retain original details that remain correct and relevant, while using the critique to correct inaccuracies. Keep the reasoning concise and straightforward.
2. Provide a confidence impact score. Rate the critique's effect on your confidence with a score from -10 to +10, where -10 means complete loss of confidence, +10 means strengthened conviction, and 0 means no impact.

Answer strictly in the following format, with no additional explanation:
Final Reasoning: <your final reasoning here>
Confidence Impact Score: <a number from -10 to 10>

STEP.D Consensus aggregation. We employ Gemini-2.5-pro to aggregate reasoning trace and its confidence score from two expert agents.

You are a gastroenterologist acting as the final arbiter. Your task is to synthesize the final reasoning trace from two experts into a single, objective conclusion. Expert's inputs:
Reasoning from Expert 1: {EXPERT.1.REASONING}
Confidence Score from Expert 1: {EXPERT.1.SCORE}
Reasoning from Expert 2: {EXPERT.2.REASONING}
Confidence Score from Expert 2: {EXPERT.2.SCORE}

Rules are as follow. Please keep the reasoning concise and straightforward.
1. Combine all points where the experts agree.
2. If the experts present opposing facts, discard those conflicting points.
3. For a point made by only one expert, check their confidence score: If > +8, discard the point. Otherwise, include it if it is substantive and objective.

Answer strictly in the following format, with no additional text or explanation:
Final Reasoning: <your final reasoning here>

STEP.E Multi-expert adjudication. Three large models – DeepSeek-R1, claude-3-haiku-20240307, and llama-3.3-70b-instruct – were utilized to perform majority voting.

You are a gastroenterologist specializing in colonoscopy, acting as the diagnostic specialist. You will receive the original question, a reasoning from other expert, and the correct answer. Based only on the information provided in the reasoning, without introducing any external knowledge or your own additional logic, judge whether the Correct Answer can be logically deduced.

Question: {QUESTION}
Reasoning: {REASONING}
Correct Answer: {REFERENCE}

Please only respond with a "YES" or "NO" response. Do not provide any additional analytical process.

D.2. Case Rejection

The proposed multi-expert debating pipeline serves not only to generate reasoning but also as a rigorous data quality control mechanism. Nearly 1/6 of samples failed to pass the final voting stage (STEP.D) and required a restart. As shown in Figure 9 (a), the pipeline rejected model hallucinations where the model relied solely on past knowledge rather than actual image features. Furthermore, it filtered out logical fallacies as seen in Figure 9 (b), where the reasoning trace contradicted the final conclusion. Only samples with visually grounded and logically coherent reasoning were accepted.

D.3. Sample Visualization

As shown in Figure 8, we visualize representative samples by selecting two sub-tasks from each of the four major clinical categories. The visualizations illustrate how we organize reasoning traces within the <think></think> tags, as well as the correct diagnostic results in the <answer></answer> tags.

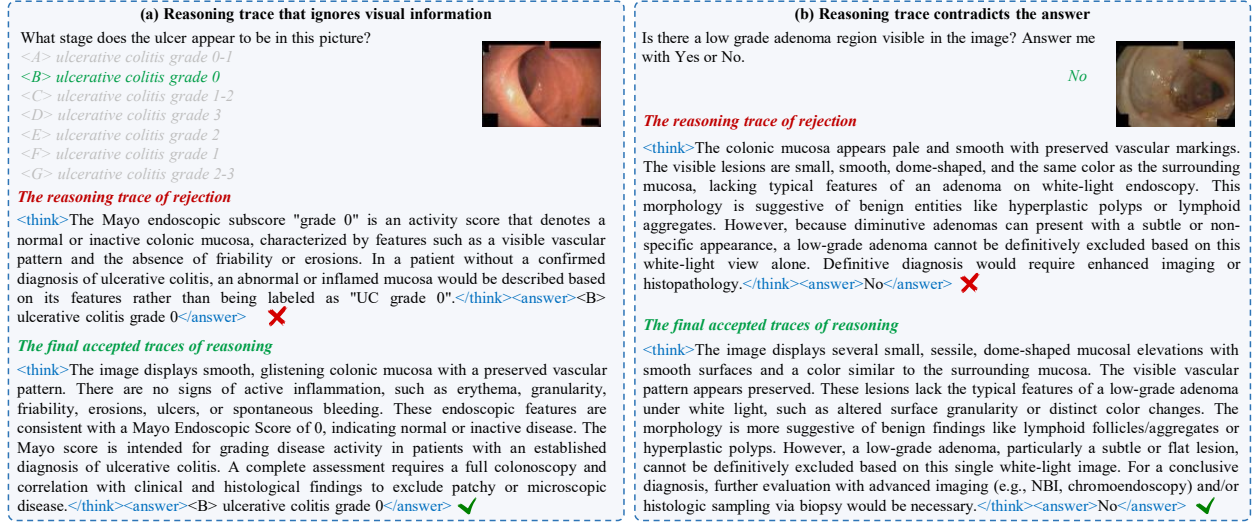


Figure 9. Examples that require restarting in our COLONREASON.



Figure 10. Qualitative comparison of COLONR1 with Med-R1 and Qwen-SFT.

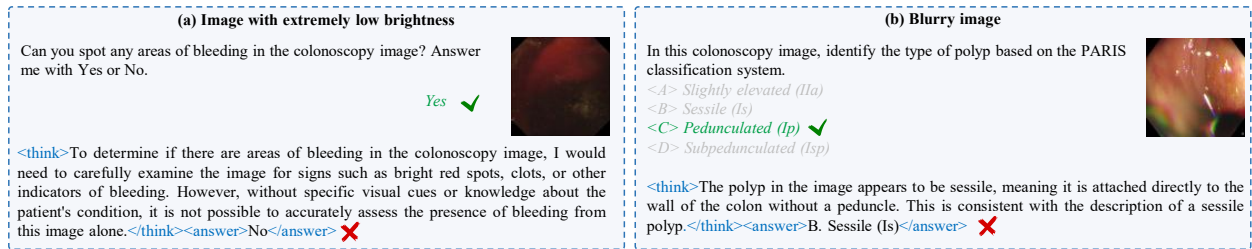


Figure 11. Two failure cases of COLONR1 on challenging scenarios.

E. Additional Details of COLONR1

E.1. Qualitative Results

Compared to the Qwen2.5VL [7] and Med-R1 [46] models, the proposed COLONR1 not only achieves higher accuracy but also exhibits a more reliable reasoning process. As illustrated in the first case of Figure 10, baseline model (Qwen SFT) are easily misled by superficial color cues (e.g., marked with red underline), incorrectly generating ‘inflammation’. In contrast, our COLONR1 captures structural features like ‘elevated’ and ‘smooth surface’ to correctly identify the low grade adenoma. Furthermore, the second case reveals a failure in visual grounding in competitors: they erroneously claim the image ‘does not provide specific details’, whereas our model successfully detects the ‘small, raised areas’ to confirm the presence of the lesion. Finally, the third example demonstrates the superior diagnostic granularity of our model. While baseline models correctly detect the object but settle for a generic label (polyp), COLONR1 goes further to infer the specific pathological type (adenoma).

E.2. Failure Cases

As shown in Figure 11, we present two failure cases caused by poor visual quality. Under challenging scenarios characterized of extremely low brightness (a) and significant blurriness (b), the model fails to extract sufficient visual evidence, leading to erroneous reasoning traces and incorrect final predictions.

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